DEVELOPMENT OF TRANSIENT STABILITY CONTROL SYSTEM (TSC SYSTEM) BASED ON ON-LINE STABILITY CALCULATION

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ABSTRACT

This paper describes a new transient stability control system named the TSC system developed for application to the trunk power system of Chubu Electric Power Co. (CEPCO). The TSC system prevents wide-area power system blackout by shedding optimal generators when a serious fault occurs.

This system has the following features:

1. **The TSC system performs detailed stability calculations based on on-line information telemetered from the actual network, and it periodically evaluates the stability of the power system against contingencies with a high degree of accuracy. The system selects the optimum generators to be shed.**
2. **If a contingency actually occurs, the generators to be shed that were determined in advance are shed about 150 ms after fault occurrence to maintain the stability of the power system.**
3. **The system can be applied to any power system configuration of CEPCO, such as a loop network, radial network, etc.**

The TSC system will be put into regular service in June 1995. It will be the world's first real-time stability control system that performs detailed transient stability calculations on a large power system on-line.

Keywords -- Power System Stability, Step-out, Generation Shedding, On-Line Stability Calculation, Screening, Shed Generation Selection, Transient Stability Control System.

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I. INTRODUCTION

The difficulty of siting power plants in suburban areas makes it necessary for CEPCO to build new power plants on sites far from the main area of consumption. This situation makes the system stability characteristics worse, and there is a possibility that a fault could extend to step-out of a group of generators causing bulk power system collapse.

Various types of power system stabilizing systems have been developed to prevent generator step-out by generation shedding. A simple one determines the generators to be shed based on conditions of a power flow and fault set in advance [1]. A complex one determines the generators to be shed based on a real-time calculation using on-line data collected before and after fault occurrence [2], [3], [4].

The conventional systems mentioned above, however, are all designed for a certain instability mode, and they cannot well be applied to a trunk power system that is subject to significant change in system stability characteristics caused by network switching, for instance, from a loop configuration to a radial configuration.

In order to cope with all power system configurations, a new transient stability control system (TSC system) has been developed. The TSC system determines the generators to be shed by performing detailed stability calculations (which are used off-line as the assessment standard of all conventional systems) based on on-line information of an actual network, and it sheds the generators when the contingency actually occurs [5]. The TSC system not only ensures system stability with high level, but also makes power system operations independent from stability problems. This means system planners and system operators can design various system configurations without reducing power supply reliability.

One of the reasons why there has been no commercially useful system that performs on-line detailed stability calculations is a lot of time consumption. To calculate system stability with an adequate level of accuracy, it takes a few minutes for only one contingency using presently available process computers. The TSC system must evaluate stability for about 100 contingencies within 5 minutes. This requirement is satisfied by efficient processing algorithms and parallel processing by multiprocessors. This paper presents the configuration, algorithms, and results of verification of the TSC system. The TSC system will be the world's first real-time stability control system that performs detailed transient stability calculations on a large power system on-line.

II. SYSTEM CONFIGURATION AND FUNCTIONS

Most cases of instability in the trunk power system of CEPCO are first swing instability. To cope with this type of instability, two to three generators have to be shed within 200 ms after fault occurrence. With the present level of technology, however, it is impossible to perform detailed stability calculations and to control generators within this required time.

The TSC system divides the process required to maintain the power system stability into two stages. The first stage is the pre-processing calculations and the second stage is the post-control process. As shown in Figure 1, the TSC system consists of three types of units; central processing unit (TSC-P) for pre-processing, fault detection equipment (TSC-C), and generation shedding equipment (TSC-T) for post-control. The TSC-P determines the optimal generators to be shed every 5 minutes by performing detailed stability calculations. The TSC-Cs register the generators to be shed against contingencies based on the TSC-P calculation results, and send generation shedding signals to the TSC-Ts when one of the contingencies actually occurs. The TSC-Ts shed the generators by the signals transmitted from the TSC-Cs. The TSC system can satisfy both high accuracy and speedy control required in the power system stabilizing system.

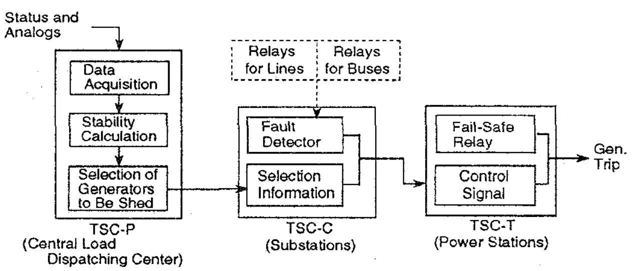


Figure 1 TSC System Configuration

The specifications of the TSC system are shown in Table 1. The scale of the power system, number of contingencies, number of equipment units are applicable to the future power system of CEPCO. Contingencies against transmission line faults are typically simultaneous double-circuit faults on the same transmission route. Bus-bar fault is calculated on the severest fault condition (3LG) to reduce the number of contingencies. Detailed stability calculations, which simulate the system dynamics, are performed with high accuracy. The generators, and their controllers are represented by detailed models, and system dynamics are calculated with an integration interval of 10 ms using the fourth-order Range-Kutta method. The TSC-P performs these calculations every 5 minutes because changes of system stability characteristic can be ignored within that time. Control completion time is designed to be about 150 ms thanks to the microwave telecommunications network owned by CEPCO.

III. PROCESSING ALGORITHM

Figure 2 shows the TSC-P processing flow. The function of each block is described below.

(B1) State Estimation:
Using measurements telemetered on-line, the current state of the power system is determined by state estimation. The weighted least squares estimation method is used in the TSC system[6].

(B2) Screening:
A preliminary evaluation of stability is made on contingencies using a fast stability evaluation method. If a contingency is judged to be stable, the detailed stability calculation is omitted.

(B3) Detailed stability calculation:
Detailed stability calculations are performed for the contingencies evaluated as unstable during screening.

Table 1 Specifications of the TSC System

Power System Scale	Generators: 100 maximum Lines: CEPCO's 500-kV and 275-kV lines and part of 154-kV lines
No. of TSC units	TSC-P: 1; TSC-C: 20 maximum; TSC-T: 100 maximum
No. of contingencies	100 maximum
Contingency conditions	Transmission lines: Radial network configuration → 1-phase 2LG, 2-phase 3LG, 3-phase 3LG 3-phase 4LG (fault duration time: 50 ms) Loop network configuration → 3-phase 6LG (fault duration time: 50 ms) Bus: 3-phase 3LG (fault duration time: 50 ms)
Detailed stability calculation	Network: Admittance matrix, Pi type equivalent circuit of positive and zero sequence networks Generator: Park model, with saturation characteristics Exciter: Detailed model with PSS (AVR: 8 types; PSS: 3 types) Governor: Typical models for hydroelectric, thermal, and nuclear power plants (3 types) Load: Constant Z, constant current, constant power characteristics Integration: Fourth-order Runge-Kutta method Integration interval: 10 ms Simulation time: 5 seconds (to 10 seconds maximum)
TSC-P calculation period	About 5 minutes (variable)
Control completion time	About 150 ms (sum of times for fault detection, transmission of shedding signal, and gen. tripping)

(B4) Stability judgement:
Using the rotor angle of each generator obtained from detailed stability calculations, stability is judged for the contingencies.

(B5) Selection of generators to be shed:
Generators to be shed are selected for the unstable cases, considering both shedding effect and operational conditions of the generators.

More details of the above (B2) Screening and (B5) Selection of generators to be shed are described below.

A. Screening Method

The screening method selects severe cases of stability at high speed among a large number of contingencies without leaving out any unstable cases. The processing flow is as follows:

(S1) A contingency is set and the value of a screening index described below is calculated.

(S2) The value of screening index is compared with the threshold value. If it is larger than the threshold value, this contingency is registered as a case that should be evaluated by detailed stability calculations. If smaller than the threshold value, this contingency is judged to be stable.

(S3) The process returns to (S1) if there is another contingency, or ends if there is none.

As described later, the TSC system is completely duplicated to enhance reliability, and each system has its own screening method. The two screening methods applied to the TSC system are discussed below. These methods are based on the stability characteristic of CEPCO's trunk network; that is, first swing instability is a typical phenomenon.

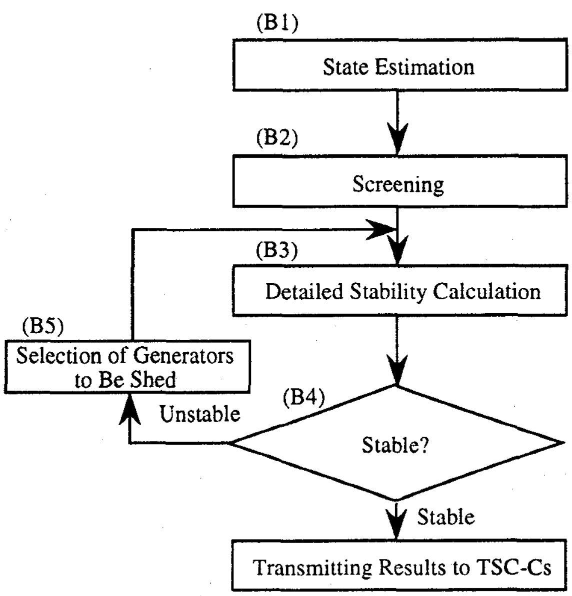


Figure 2 Processing Flow of TSC-P

1) Deceleration Power Index Method

Because the fault duration time of all contingencies is only 50 ms (3 cycles), stability will be strongly affected by a drop in synchronizing power following a change in the network configuration after fault clearance. To express the drop in a synchronizing power after fault clearance, a new stability index DP (deceleration power) is defined for the screening.

$$DP = \sum_{i=1}^{N} \{ Mi \bullet | Pdi | / (\sum_{i=1}^{N} Mi) \} \quad \text{.............................}(1)$$

where

$$Pdi = (Pci - Poi) / Mi - \sum_{i=1}^{N} (Pci - Poi) / (\sum_{i=1}^{N} Mi) \quad \text{........}(2)$$

where N: number of generators; Mi: inertia constant; Poi: generator output before fault; Pci: generator output at instant of fault clearance.

Pdi in equation (2) is a deceleration power of each generator with respect to the center of inertia frame. When a generator decelerates with respect to the center of inertia frame after a fault, Pdi becomes positive, and when a generator accelerates, Pdi becomes negative. Therefore, a contingency with a larger value of DP, which is the weighted average magnitude of Pdi, is considered a severe case for stability.

To calculate DP using equations (1) and (2), the generator output at instant of fault clearance, Pci, is required. To obtain Pci, the stability calculation must be done until the instant of fault clearance, and it takes about 1/100 of the time required for a 5-second detailed stability calculation. Therefore, screening using DP is about 100 times faster than a detailed stability calculation.

2) Simplified Stability Calculation Method

Screening is performed using a fast stability calculation program that employs several speed-up measures such as model simplifications. The maximum generator rotor angle obtained from the program is used as a screening index. The differences between the simplified program and the detailed program are shown below.

(1) Generator controllers:
AVR (Automatic Voltage Regulator) is modeled as a simplified one that ignores PSS (Power System Stabilizer). Governor turbine model is omitted because its effect on stability is rather small.

(2) Imbalance fault simulation:
Imbalance faults are simulated by inserting the driving point impedance of negative and zero sequence networks viewed from the fault point into the positive sequence network as a fault impedance. This makes it unnecessary to calculate the negative and zero sequence networks and shortens the calculation time.

(3) Integration method:
The Trapezoidal method is used because it permits faster calculations than the fourth-order Runge-Kutta method that is used for detailed stability calculations.

(4) Simulation time:
One-second simulation time is sufficient to apply to first swing instability.

The model simplification and shortened simulation time described above raise processing speed about 30 times compared with the case of a 5-second detailed stability calculation.

B. Selection of Generators to Be shed

To minimize the number of generators to be shed and/or the amount of generation shedding required for stabilizing, it is necessary to extract the stepped-out generators and select the optimum ones among them. The efficient method of determining generators to be shed that is applied to the TSC system is described below.

1) Extraction of Stepped-Out Generators

A step-out occurs when the relative angle differences between the generators spread after fault occurrence. Using the rotor angle of each generator obtained from detailed stability calculation, the generators stepped-out can be determined by equation (3).

$$(\delta i\ (t) - \delta s(t)) \geq \delta 1 \qquad (3)$$

where δi: rotor angle of each generator; δs: rotor angle of reference generator; t: simulation time in detailed stability calculation; $\delta 1$: threshold value for stability evaluation.

A certain group of stepped-out generators may affect another group, leading to system-wide instability. In this case, generators to be shed must be selected among the generators stepped-out first. Because the rotor angles of these generators increase faster than those of another generator, step-out decision time DT is applied to extracting the stepped-out generators. In more detail, the generators evaluated as unstable within DT just after the first generator is judged as unstable using equation (3) are extracted as stepped-out generators.

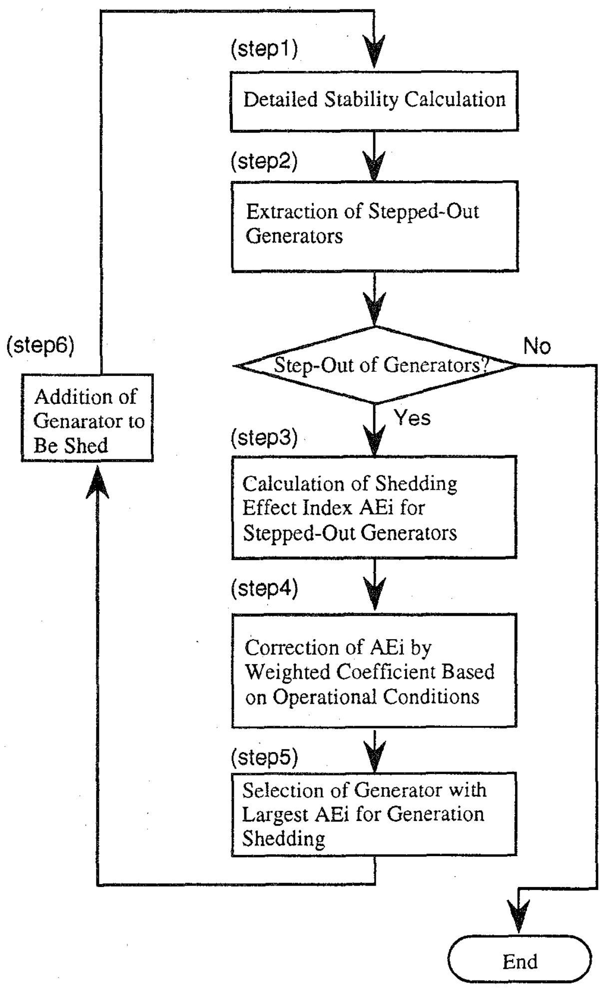


Figure 3 Selection of Generators to be Shed

2) Evaluation of Generation Shedding Effect

To select some generators to be shed from the stepped-out generators, it is necessary to quantitatively evaluate their shedding effect. One possible way, that can be called the sequential step-out method, is to select the generator that is stepped-out at first. However, this method might select first small-capacity generators that have a small inertia constant. As a result, the number of generators to be shed increases. Therefore, the acceleration energy, AEi, of each generator at the start of a step-out shown in equation (4) is used as an index of generation shedding effect. AEi is the energy that contributes to generator instability. The larger the value of AEi a generator has, the larger will be the stabilizing effect produced by generation shedding. This is called the AE method.

$$AEi = \frac{1}{2} Mi\ (\omega i(tc) - \omega s(tc))^2 \qquad (4)$$

where ωi: rotor speed of each generator; ωs: rotor speed of reference generator; tc: time when a generator is judged as having been stepped-out first by equation (3); Mi: inertia constant.

3) Flow of Selection of Generators to Be Shed

The TSC system is designed to cover instability modes in which a few to ten-odd generators are stepped-out, and one to a few generators must be shed for stabilization among ten-odd stepped-out generators, There are more than 100 possible combinations to select generators to be shed, and it is difficult to determine the best solution at once. The solution to this problem found for the TSC system is to select generators to be shed by repetition of detailed stability calculations. This method selects the n-th generator to be shed, using the AEi obtained from detailed stability calculation results at the time of shedding the (n-1) generators.

Figure 3 shows the flow of selection of generators to be shed discussed so far. Step 4 in Figure 3 is the processing to take into account the operational conditions of generators. In Step 4, the index of generation shedding effect, AEi calculated in Step 3, is multiplied by the weight coefficient peculiar to the generator to take into account the operational conditions. For example, a weight coefficient of 1.1 may be set for an FCB (fast cut back) generator that can be restarted in a short time after shedding.

As shown in the processing flow of Figure 3, generators with a high shedding effect can be efficiently selected one by one by repeating detailed stability calculations, taking into account the operational conditions of generators. It leads to the fact that if n generators are to be shed for a contingency, (n+1) detailed stability calculations are required.

IV. ALGORITHM VERIFICATION USING TSC MODEL

A. Test Configuration

To evaluate the validity of the screening method and selection method of generators to be shed discussed above, a prototype model (called the TSC model hereafter), with a processing algorithm equivalent to the TSC system, was provided. It consists of two workstations, each with a different screening method. The TSC model is connected to an analog simulator via a gateway as shown in Figure 4 to conduct a verification test. The analog simulator consists of analog models of generators, loads, transmission lines, etc., a host computer to manage the system, and workstations to make out a simulated power system [7]. In this test configuration, the analog simulator corresponds to the actual power system and the TSC model to a TSC-P. Using the analog simulator, tests can be conducted on future power systems and power systems with severe stability conditions.

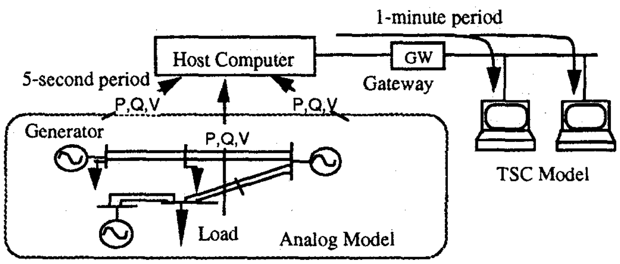


Figure 4 Test Configuration Using TSC Model and Analog Simulator

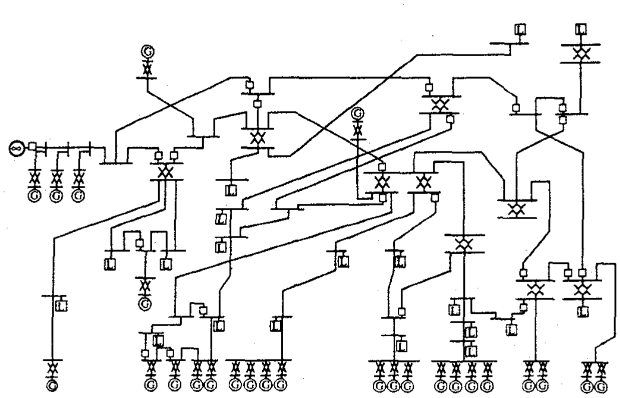


Figure 5 Example of Network Configuration

An example of network configurations on which a verification test was conducted is shown in Figure 5. The verification procedures are shown below.

(1) Simulated power system data, including the line constants and generator constants created on the screen of a workstation of the analog simulator, are transferred from the host computer to the TSC model.

(2) By the analog simulator the power consumption of the load is adjusted at the rate of about 1 percent per minute to gradually change the state of the power system in the direction of increasing severity of stability.

(3) Active power flows and other data measured in the analog model are transmitted from the host computer to the TSC model every minute. After a series of processes of the TSC model, swing curves of generators and numeric values serving as the basis of selecting generators to be shed are displayed and analyzed.

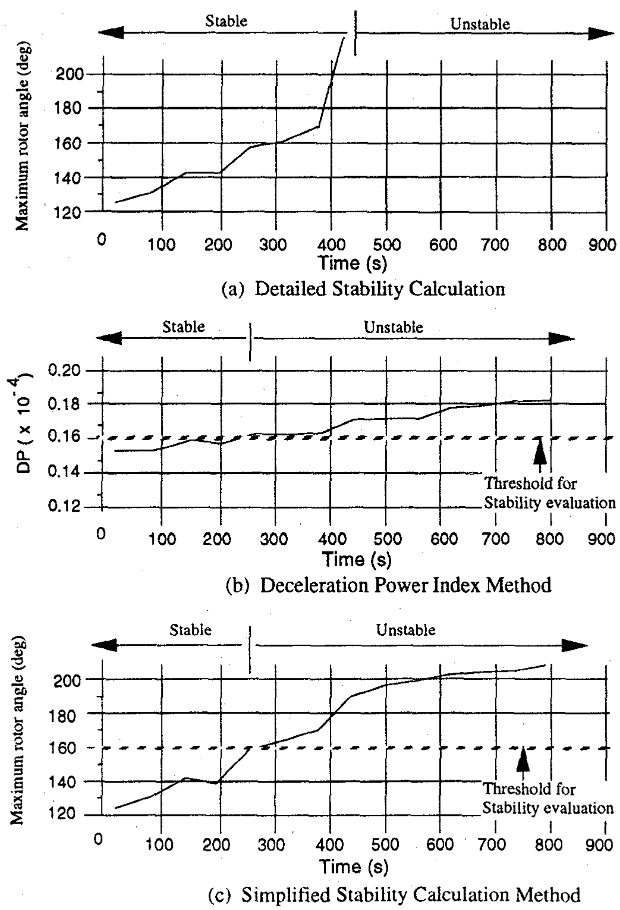


Figure 6 Comparison Between Dtailed Stability Calculation and Screening Method

B. Test Results

The power system simulated by the analog simulator was the network of the Chubu Electric Power Company over the period of 1995 to 2000. By changing network configurations, power flow conditions, and contingencies, about 2000 cases were examined for theoretical verification.

An example of comparison between the results of detailed stability calculations and the screening methods is shown in Figure 6. The contingency considered is a 3-phase-4LG fault on double-circuit transmission lines. Figure 6(a), (b), and (c) show the maximum angle over 5-second simulation by detailed stability calculations, the DP in the deceleration power index method, and the maximum generator rotor angle in the simplified stability calculation method. The horizontal axes of these figures represent the simulation time on the analog simulator.

From Figure 6(a), the stability of the simulated power system is becoming severer with time, and generators are stepped-out at the instant of 450 seconds. Figure 6(b) and (c) show that the relations between the screening indixes and stability, that is, the result of detailed stability calculations, have positive correlations. This means it is possible to set a threshold value for each screening method that is adequate to avoid misclassification of unstable cases as stable. For example, if the threshold value for stability evaluation is set as indicated by the dotted lines in Figure 6(b) and (c), both screening methods can judge that the simulated power system is unstable for the contingency at instant of 250 seconds. It means that they have safe judgments. The results described so far verify the effectiveness of the proposed screening methods.

The verification results of the selection method of generators to be shed are shown in Figure 7. To verify the validity of the AE method based on the generation shedding effect index discussed in Chapter Ⅲ, the AE method was compared with the sequential step-out method, which selects the generators to be shed based only on the order in which their rotor angles exceed the specified threshold value. Figure 7(a) shows a comparison of the two methods in terms of shedding quantity, and Figure 7(b) shows a comparison of the same in terms of the number of generators shed. The AE method tends to select a large-capacity generator first and the shedding quantity of the first selected generator is somewhat larger compared with the sequential step-out method, but the expected value of shedding quantity is about the same as that of the latter. On the other hand, the AE method sheds fewer generators than the sequential step-out method and thus is advantageous in restoring the power system to the normal condition. In the actual system, therefore, the AE method is superior to the sequential step-out method.

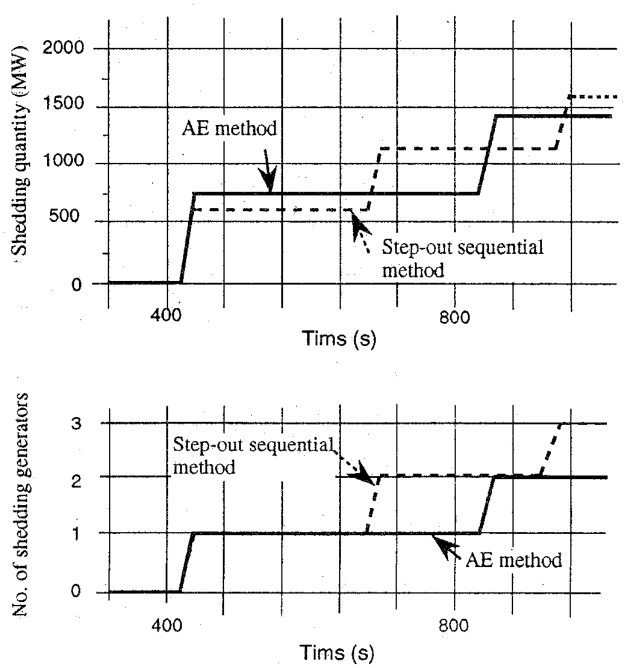


Figure 7 Verification Results of Selection Method of Generators to Be Shed

V. COMMERCIAL TSC SYSTEM AND FIELD TEST RESULTS

A. Commercial TSC System

Figure 8 shows the configuration of TSC-P. It is a dual computer system, consisting of a primary and a standby subsystems. Each subsystem consists of a management processor and arithmetic processors. The arithmetic processors process screening, detailed stability calculation, and selection of generators to be shed -- which demands the longest processing time of all the processes performed by TSC-P -- in parallel in blocks of contingencies. Information concerning generators to be shed is transmitted to TSC-Cs via the communications network. This information is also transmitted to the central load dispatching center for use in power system operation. Each subsystem is provided with a workstation for maintenance and test purposes.

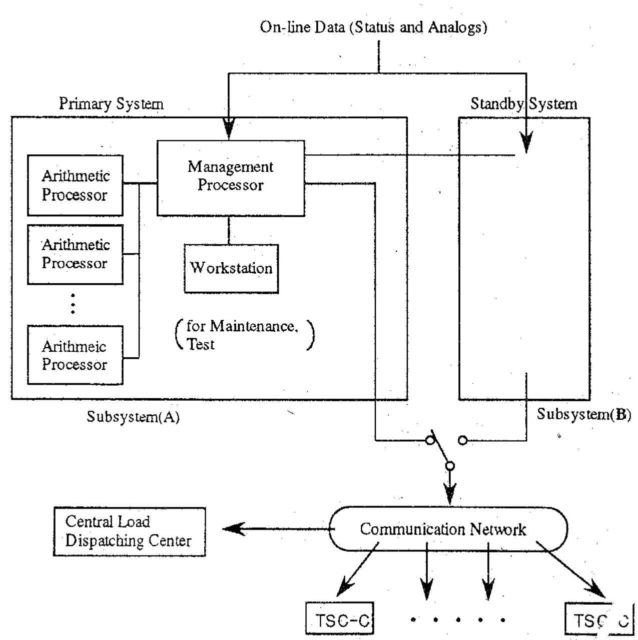
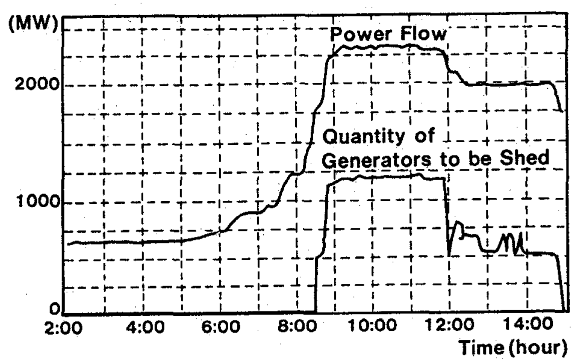


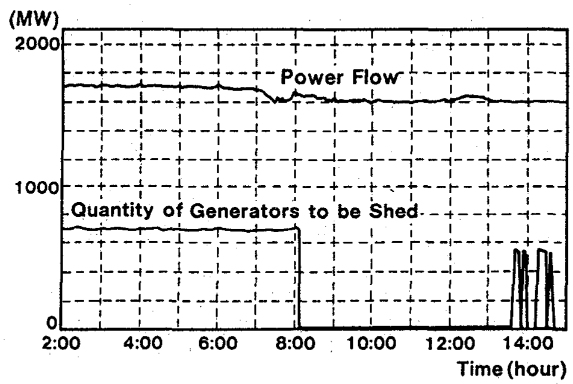
Figure 8 Configuration of Commercial TSC System

To avoid failure to operate, the commercial TSC system has the following countermeasures:

(1) One subsystem employs the DP method and the other employs the simplified stability calculation method for screening.
(2) Generators selected by the standby system to be shed are transmitted to the primary system via a communications line. The numbers of generators to be shed obtained from both subsystems are compared, and the larger one is selected as the final result to be transmitted to the TSC-Cs.



(a) Fault near Thermal Plant with Large Output Regulation



(b) Fault near Thermal Plant with Small Output Regulation

Figure 9 Change in Shedding Quantity and Power Flow

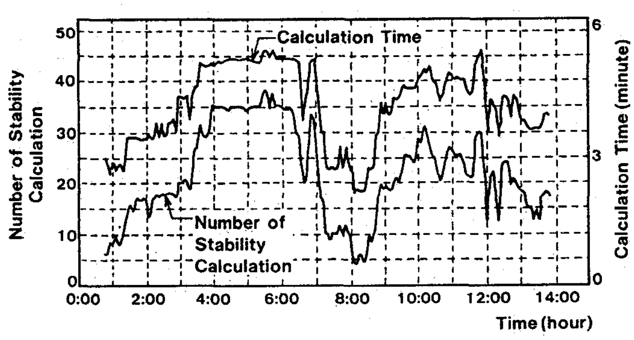


Figure 10 Transition of Calculation Time and Number of Stability Calculations

(3) The threshold values for screening are tuned automatically using the relation between results of detailed stability calculations and values of the screening indexes.

These countermeasures prevent the screening from misclassifying an unstable case as stable.

B. Field Test Results

A field verification test was conducted on the system in August 1994 on the basis of the actual on-line information. A total 57 contingencies involving transmission line faults and buse faults were covered.

An example of the field verification test is shown in Figure 9. Figure 9 shows the change in the quantity of generators to be shed and pre-fault power flow of the lines, on which contingencies may be applied, over the period of 13-hours. Contingencies are two 3-phase-4LG faults on double-circuit transmission lines close to thermal power plants. Figure 9(a) shows the result of the fault near one thermal power plant whose output varies largely over the day, and Figure 9(b) shows the same result for the other thermal power plant whose output is almost constant.

As shown in Figure 9, the former plant is stable during the night, but with increase of the generator output to meet demand in the morning, stability is sharply aggravated. In contrast, the latter plant tends to encounter growing severity of stability problems at night. It is also confirmed that relationship between change in quantity of generators to be shed and change in power flow of the lines shows a considerable strong correlation in the case of the fault near the former plant. But these is no clear correlation between them in the case of the fault near the latter plant. It is evident from these results that it is difficult for the conventional system, which determines the quantity of generators to be shed based solely on the power flow and fault condition, to determine appropriate conditions of generation shedding, particularly for faults near a thermal power plants such as the latter one.

C. Processing Time

During the field verification test mentioned above, processing time was also measured. Breakdown of the processing time is shown in Table 2. Detailed stability calculations of 30 cases were completed in about 5 minutes as originally designed. The transition of calculation time and number of detailed stability calculations are shown in Figure 10. Figure 10 shows that there

Table 2 Processing Time of TSC-P

Processings	Time	Remarks
State Estimation (including On-line Data Collection and Network Model Build)	1 min. 8 sec.	
Screening	19 sec.	57 cases
Detailed Stability Calculation and Selection of Generators to Be shed	3 min. 15 sec.	30 Detailed Stability Calculations
Others (including Transmitting Results to TSC-Cs)	24 sec.	
Total	5 min. 6 sec.	

is a positive correlation between the processing time and the number of detailed stability calculations. This is presumably because the processors have little waiting time for processing owing to proper task allocation.

VI. Conclusions

The new transient stability control (TSC) system based on on-line transient stability calculations were presented in this paper. The high-speed, efficient processing algorithm that is applied to the TSC system and the results of verification tests were also described. The features of the TSC system are as follows:

- The TSC system performs detailed stability calculations based on on-line information of the actual power system and evaluates the stability against contingencies with a high degree of accuracy.
- The system automatically selects generators to be shed as best suited to maintain stability.
- If a contingency actually occurs, the generators determined to be shed by precalculation are shed about 150 ms after fault occurrence.
- The system can be applied to any power system configuration of CEPCO, such as a loop network, radial network, etc.

The TSC system can prevent the power system from wide-area blackouts by shedding generators optimally when a serious fault occurs. It will be put in regular service in June, 1995. Transient stability of a large power system within 5 to 10 seconds can now be monitored and controlled by this TSC system. Dynamic stability will be our next subject.

VII. Acknowledgments

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Discussion

Nicholas W. Miller, Einar Larsen, Satoru Ihara: The authors are to be complemented on reporting a truly remarkable advancement in the state-of-the-art in on-line control. The system combines considerable practical insight with the high-performance computational ability of modern equipment to provide a valuable operations tool.

Can the authors elaborate a little more on the benefits of this system? We presume having this TSC in place will permit more aggressive loading of the transmission system. Specifically, could the authors address how they see this system as affecting the following aspects of operation:

- Spinning reserve margins
- Degree of looping within transmission system
- Degree of interchange with neighboring utilities

The use of the Deceleration Power Index Method seems to work nicely for a tightly interconnected, high short circuit strength system such as the Chubu power system. Do the authors have any insights as to which types of systems this method will have more benefit than others? For example, a widely dispersed system, with long transmission corridors, such as that found in the western portion of North America would be expected to have considerably different characteristics.

The authors state that the only issue is first-swing transient stability, yet simulations are run to 5 seconds. What is the typical and longest time from the simulated disturbance to the time a case is deemed unstable? Is this likely to change as the system is operated closer to its limit in the future?

The overall structure of this decision structure would lend itself well to other switching decisions as well. One technique which is used in the western US is that of controlled islanding. It seems that this technique could be adapted quite successfully to the implementation of islanding schemes. Do the authors have any insights which may help others considering such applications?

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Y. Kitayama, Y. Kokai, K. Omata, N. Fukushima: The authors wish to thank the discussers for their interest in the paper and their comments. Since a number of comments and points are raised, we would like to respond to each of them by numbering them as follows:

1. Benefits of the system

1.1 Spinning reserve margins

The whole 60-Hz power system of Japan including the Chubu Electric Power Company (CEPCO) network has a supply capacity of about 100 GW as of the 1995 summer, of which 3 to 5% is considered to be spinning reserve. On the other hand the total output power of generators to be shed by the TSC is 3 GW at most. So, there is no need to change the spinning reserve margins at present.

1.2 Degree of looping within the CEPCO network

The looped system where power flow largely changes due to power inflow and outflow could hardly be controlled by the conventional type of TSC which determines generators to be shed only by comparing some power flows with the pre-set values. Since the newly developed TSC performs stability calculations on-line, it can determine generators to be shed as required whatever the power system configuration may be. In this sense, the CEPCO network can now be easily looped to any degree with the TSC. Generally speaking, a looped network is more stable than any radial network in terms of transient stability. Therefore the TSC makes it possible to choose a safer system configuration.

1.3 Degree of interchange with neighboring utilities

At present there is only one neighboring utility linked via AC transmission lines to the CEPCO network. Furthermore, CEPCO is always importing power from the neighboring utility except for emergencies. So, there is little direct impact on degree of interchange with neighboring utilities by the TSC, now. If CEPCO will have a sufficient power generation capacity to export power to neighboring utilities in a future, however, the newly developed TSC will offer a great advantage in enabling CEPCO to increase its power export up to the thermal capacity of the tie lines.

2. Deceleration Power (DP) Index Method

Generally, the transient stability of a power system is subject to the following three factors:

(1) Power consumption by the loads and generator output conditions before fault occurrence

(2) The extent of imbalance in acceleration energy among the generators during the fault

(3) Loss of synchronizing capacity incidental to the change in the network configuration after fault clearance.

DP is mainly an index representing the effect of stability regarding the factor (3) above. There are the following two reasons for the effective application of the DP Index Method to the CEPCO power network.

(1) Faults last short, only 50 ms, so their effect on the stability of generator acceleration energy during a fault is rather small.

(2) Fault mode consists mainly of unbalance faults over two lines, and multi-phase shedding incidental to fault clearance sharply lowers the transmission capacity.

Therefore, the applicability of the DP Index Method depends on the stability characteristics of the power network, rather than on its system configuration. In order words, in case of power networks whose stability is adversely affected by the lowering of synchronizing capacity after fault clearance, the DP Index Method can be applied to them regardless of its system configuration.

3. Simulation Time

A decision on instability is made if there is any generator whose internal phase difference angle has exceeded the threshold level. If the threshold level is 360° (in reference to the generator serving as the system center), a decision on instability is made in about 1.5 seconds after fault occurrence in most cases.

The maximum time it may take to make a decision on instability will be about 2 seconds, considering the first swing instability only.

5 seconds is specified in TSC simulations, which however are naturally suspended upon a decision on instability.

At stability limits, there are cases where the generators will step out not on the first swing but on the second or third swing. Therefore, simulation time is specified to be normally 5 seconds, which may be extended up to 10 seconds.

4. Controlled Islanding Scheme

The authors fully agree that a system like the TSC which performs stability calculations on-line can be applied to some other schemes. We, however, are not familiar with controlled islanding scheme suggested by the discussers. So, the following are our comments only:

Indeed, the TSC system can perform simulations on any independent power system, and determine whether it is stable or unstable against faults or switching operations.

For the safest possible operation of power networks, however, it is necessary to develop a theory which would indicate which point of tie lines should be disconnected. But, it is still unknown to us.

Because of better spinning reserve margins, easier frequency control, etc., in Japan, the convenience of power interchange with neighboring networks is far better appreciated than the advantage of the independent power system, and therefore the suggested controlled islanding scheme will have little opportunity of being accepted here, if stability collapse can be positively avoided by generation shedding. We think, however, a suggested controlled islanding scheme with a control system like the TSC can be applied in a different environment where many of small power systems are heavily interconnected with each other (e.g., in North America).

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